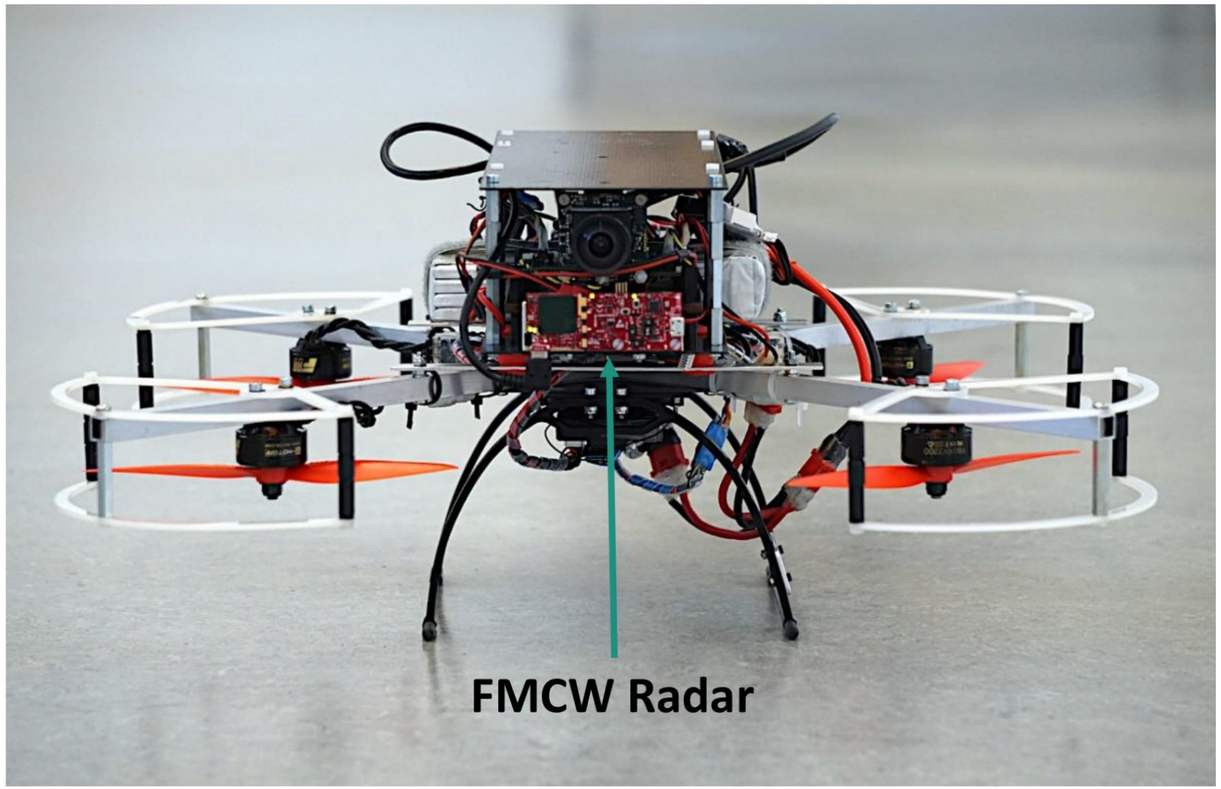
# CREVE: An Acceleration-based Constraint Approach for Robust Radar Ego-Velocity Estimation

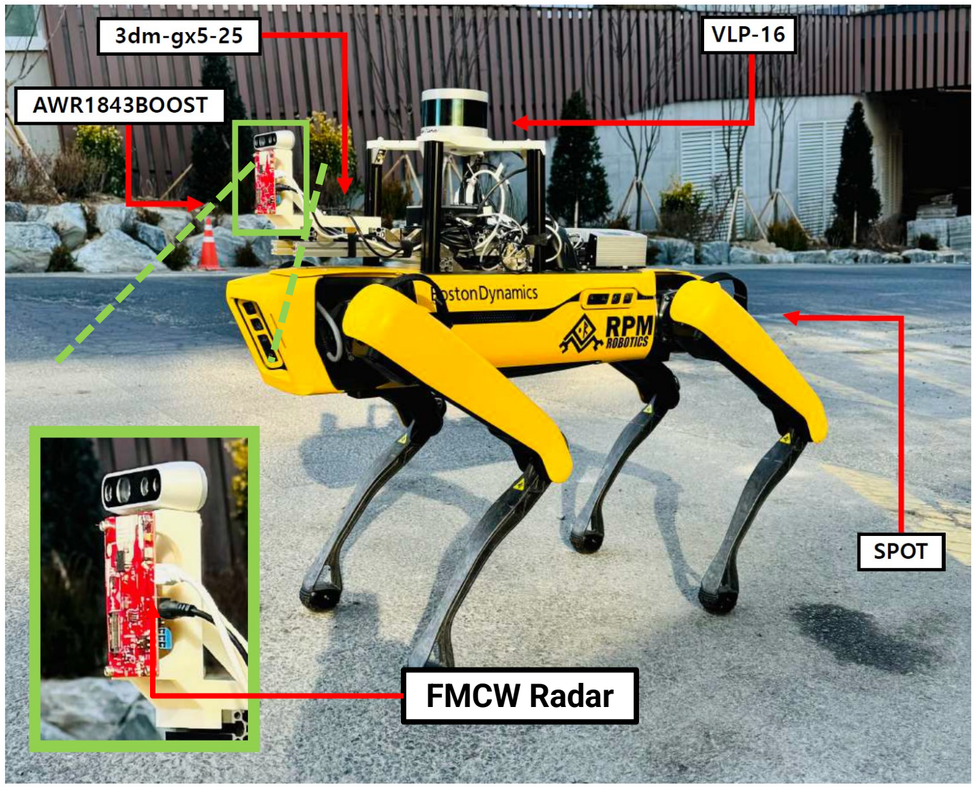
Hoang Viet Do, Bo Sung Ko, and Jin Woo Song

Online at: https://arxiv.org/abs/2409.16847

# Introduction (1/3) – FMCW Radar

- mmWave FMCW Radar has gained significant attention.
  - Low cost, compact size, and all-weather operation.



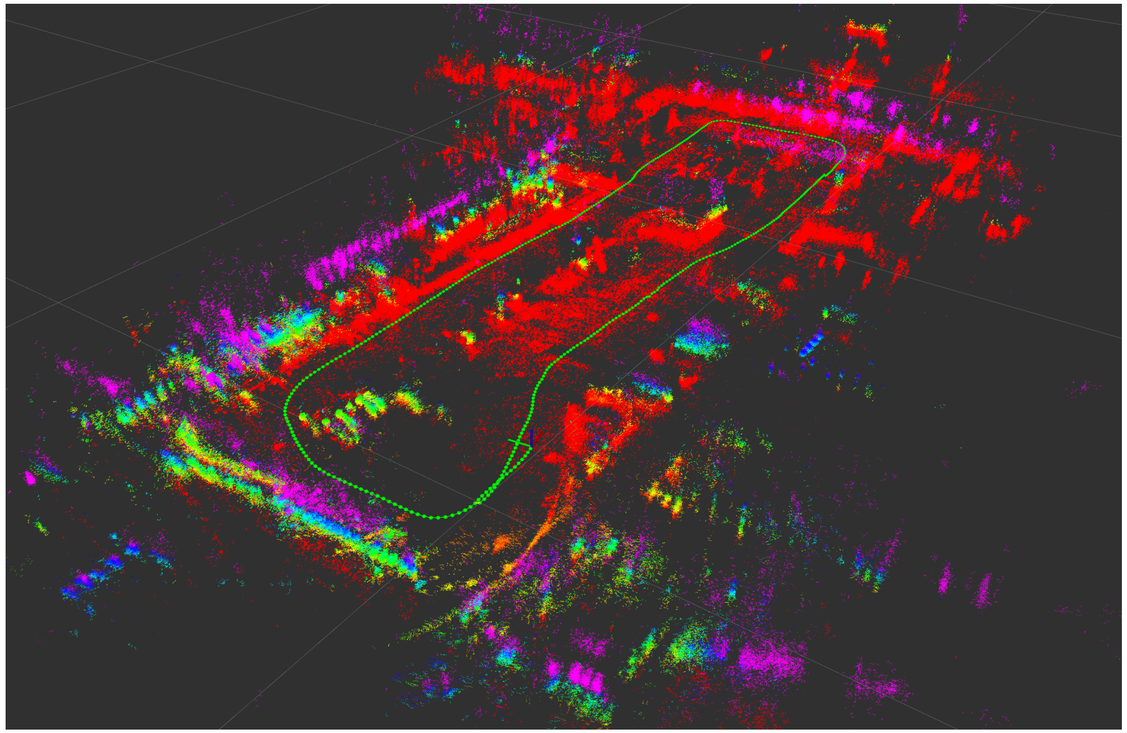



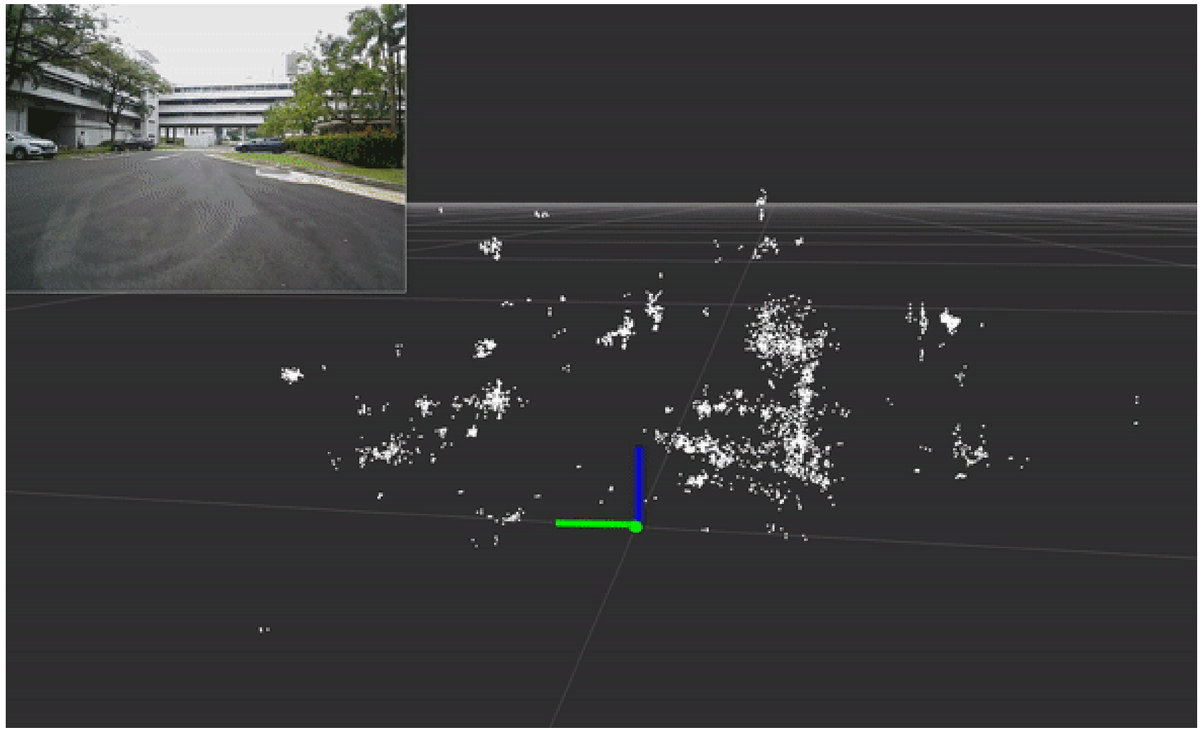

C. Doer and G. F. Trommer, "Radar Inertial Odometry With Online Calibration," *2020 European Navigation Conference (ENC)*, Dresden, Germany, 2020, pp. 1-10, doi: 10.23919/ENC48637.2020.9317343.

S. Jung, W. Yang, and A. Kim, “Co-ral: Complementary radar-leg odometry with 4-dof optimization and rolling contact,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*., to be published.

# Introduction (2/3) – Application

- Promising alternative to LiDAR and Camera for SLAM.
  - LiDAR and Camera are often limited by poor lighting, illumination, and environment factors.

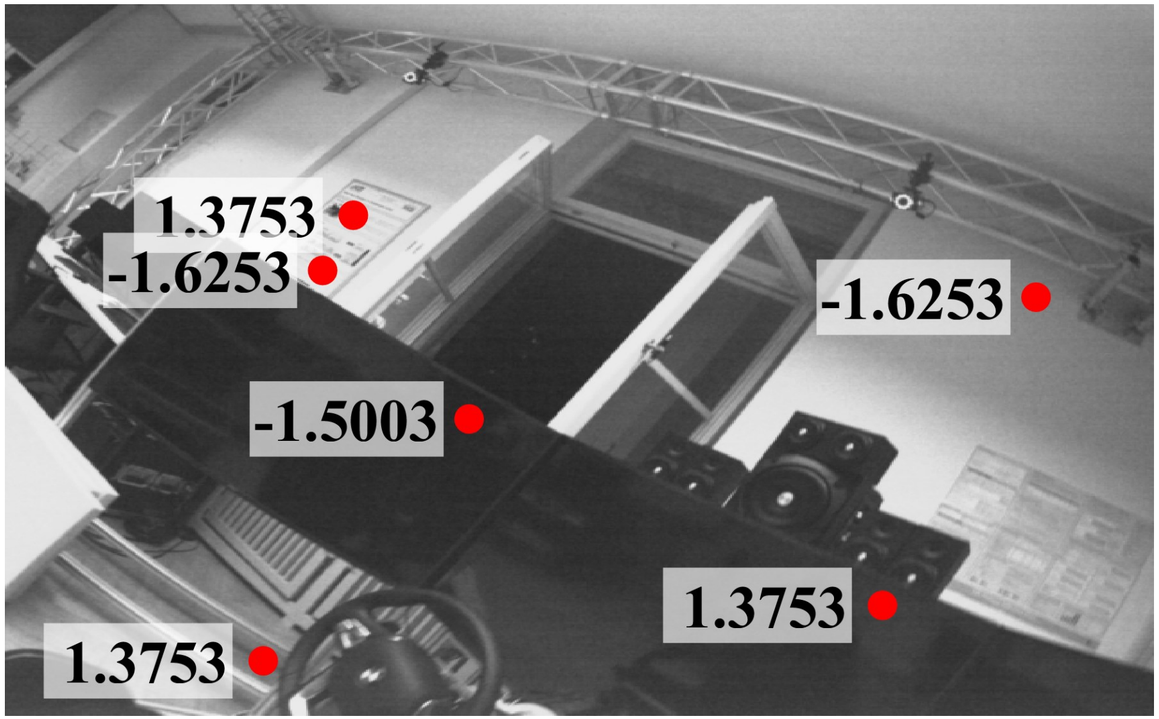


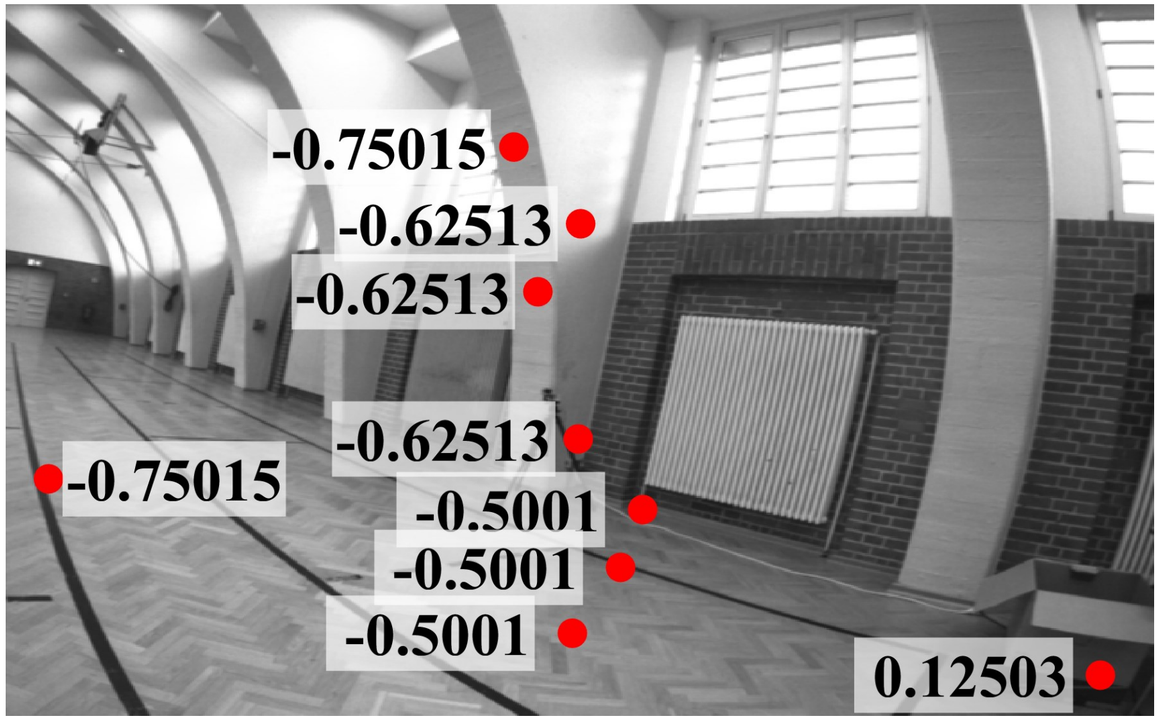


J. Zhang, H. Zhuge, Z. Wu, G. Peng, M. Wen, Y. Liu, and D. Wang, “4dradarslam: A 4d imaging radar slam system for large-scale environments based on pose graph optimization,” *in 2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 8333-8340.

# Introduction (3/3) – Radar Point Cloud

- Radar point clouds are known to be noisy and sparse due to ghost targets and multipath signals.
  - Typical commercial radar provides 4D point cloud (range, bearing and Doppler velocity). $P = [p_x^r \quad p_y^r \quad p_z^r \quad v_D]$




C. Doer and G. F. Trommer, "Radar visual inertial odometry and radar thermal inertial odometry: Robust navigation even in challenging visual conditions," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 331–338.

# Prior Works (1/3) – REVE (Traditional)

- Relationship between 3D body velocity $v_k^r$, target Doppler velocity $v_D$ and target normalized 3D position $\bar{p}$.
- REVE ≙ RANSAC/LSQ.
  - Using least square (LSQ), we can estimate $\hat{v}_k^r$.
  - Combining with RANSAC, an outlier-robustness estimate can be obtained.

$$\underbrace{\begin{bmatrix} -v_{D,1} \\ -v_{D,2} \\ \vdots \\ -v_{D,N} \end{bmatrix} = \begin{bmatrix} \bar{p}_{x,1} & \bar{p}_{y,1} & \bar{p}_{z,1} \\ \bar{p}_{x,2} & \bar{p}_{y,2} & \bar{p}_{z,2} \\ \vdots & \vdots & \vdots \\ \bar{p}_{x,N} & \bar{p}_{y,N} & \bar{p}_{z,N} \end{bmatrix} \begin{bmatrix} v_x^r \\ v_y^r \\ v_z^r \end{bmatrix}}_{y_k = H_k v_k^r}$$

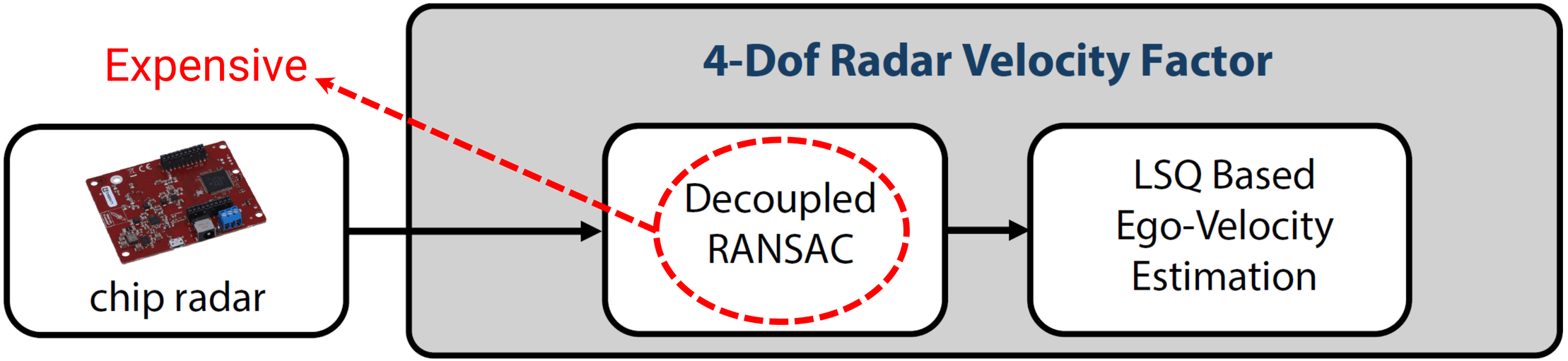
LSQ Solution:

$$\hat{v}_k^r = \left(H_k^\top H_k\right)^{-1} H_k^\top y_k$$

C. Doer and G. F. Trommer, “An ekf based approach to radar inertial odometry,” in *2020 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, 2020, pp. 152–159.

# Prior Works (2/3) – DeREVE

- REVE has been said to be sensitive and provide inaccurate ego-velocity estimation in z−axis.
- DeREVE ≙ Decoupled REVE.
  - Cascade RANSAC/LSQ.
  - First, RANSAC/LSQ for only $x-y$ plane → Inlier set with $x'$ and $\hat{v}_{xy}^{r}$.
  - Second, RANSAC/LSQ for $x'-\mathrm{z}$ plane with previous $\hat{v}_{x}^{r} \rightarrow \hat{v}_{x'z}^{r}$.
  - Last, refinement using only LSQ to get 3D ego-velocity $\hat{v}_{k}^{r}$.




S. Jung, W. Yang, and A. Kim, “Co-ral: Complementary radar-leg odometry with 4-dof optimization and rolling contact,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*., to be published.

# Prior Works (3/3) – RAVE

- Using a set (window) of $N$ previous estimated ego-velocity.
  - If the difference between the current estimate and the average norm of velocity estimates in the window is bigger than a pre-defined threshold, then reject the current estimate.

**Algorithm 1** The RAVE filter

**Inputs:** Sliding windows size $N$, velocity estimates $\mathbf{v}_{k-(N+1):k}$, thresholds $T_1$, $a_{max}$

**Output:** Updated window of last $N$ valid velocity estimates

1: Calculate the average norm of velocity estimates in the window $\mathrm{n}_{\mathrm{avg}} = \frac{1}{N}\sum_{i=1}^{N}\|\mathbf{v}_{k-i}\|$

2: **if** $|\mathrm{n}_{\mathrm{avg}} - \|\mathbf{v}_k\|| < T_1$ and $\|\mathbf{v}_k - \mathbf{v}_{k-1}\|/(t_k - t_{k-1}) < a_{max}$ **then**

3: Estimated velocity is valid: drop $\mathbf{v}_{k-(N+1)}$ and add $\mathbf{v}_k$ to the window

4: **else**

5: Estimated velocity is invalid: Reject $\mathbf{v}_k$

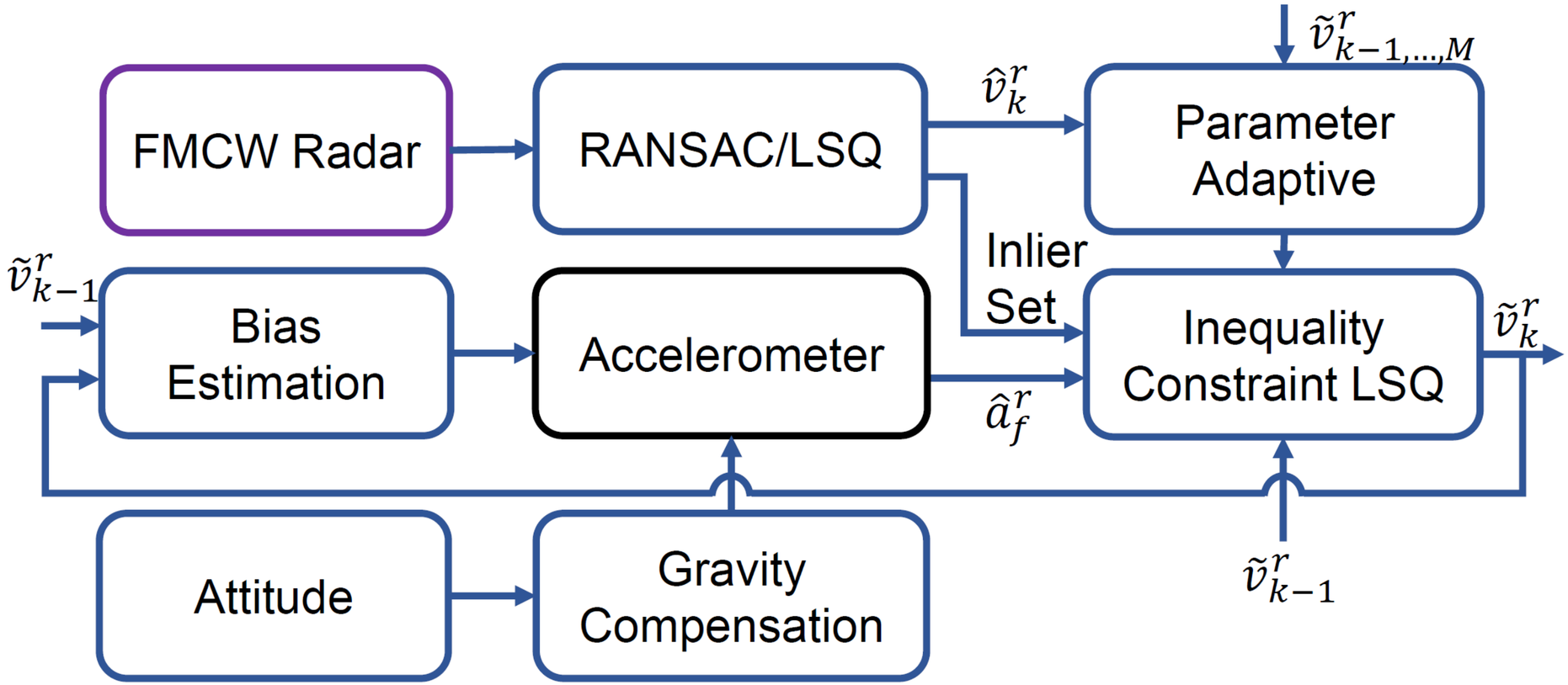
6: **end if**

Reduce number of measurement

V. -J. Štironja, L. Petrović, J. Peršić, I. Marković and I. Petrović, "RAVE: A Framework for Radar Ego-Velocity Estimation," *2024 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, Pilsen, Czech Republic, 2024, pp. 1-6.

# Methodology (1/5) – Overview

- We propose an acceleration-based **C**onstraint method for robust **R**adar **E**go- **V**elocity **E**stimation, termed CREVE.
  - Accelerometer bias estimation.
  - Parameter adaptation rule for adjusting inequality constraint range.



# Methodology (2/5) – Constraint

- Given the rotation matrix and accelerometer bias estimation $\hat{b}_a$ at time instance $k$, the estimated acceleration expressed in the radar frame $\hat{a}_f^r$ can be calculated as

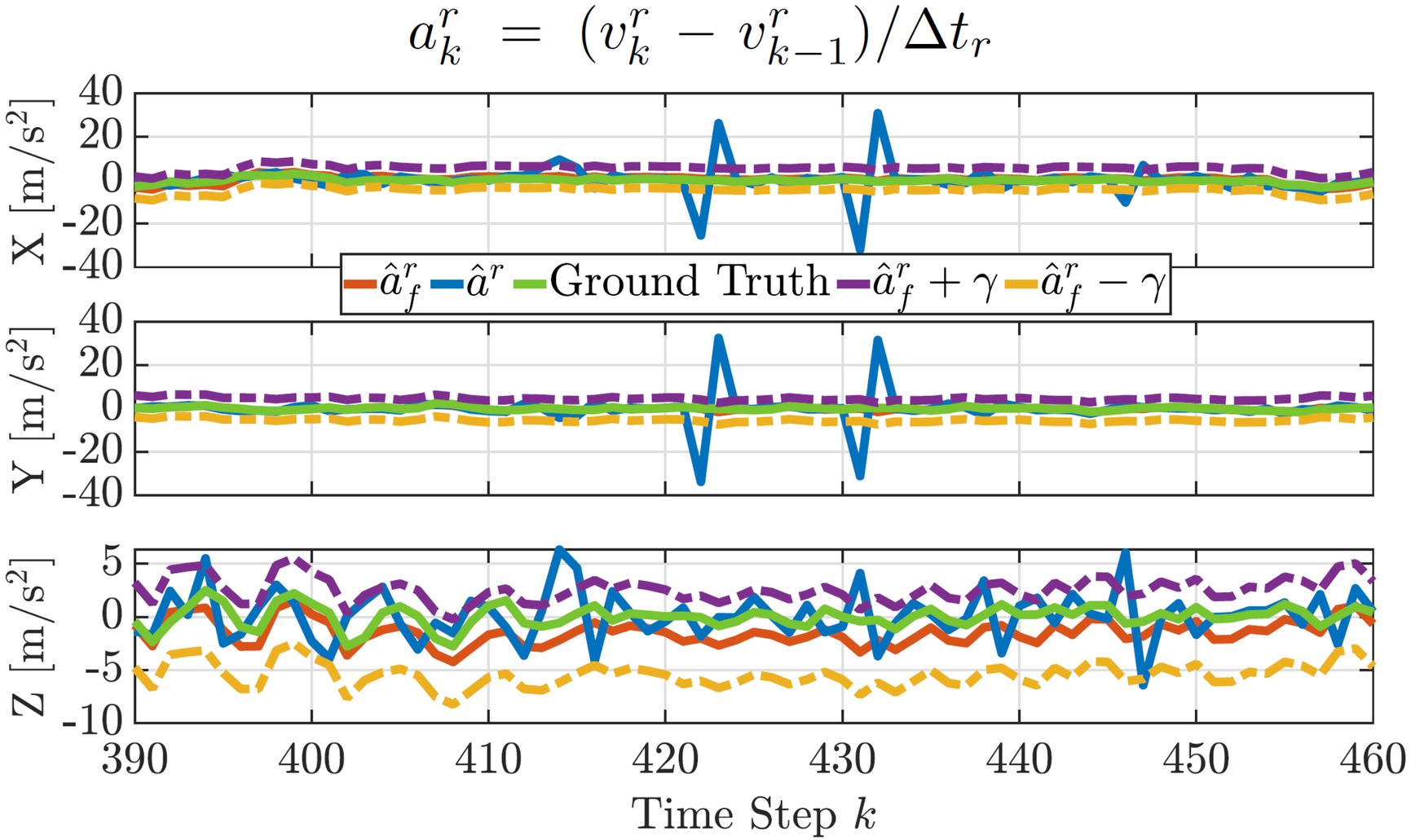
$$\hat{a}_f^r = C_b^r \left( \bar{f}^b - \hat{b}_a + C_n^b g^n \right)$$

- Based on the above equation, we formulate the following linear programming (LP) problem

$$\begin{aligned} \underset{\tilde{v}_k^r \in \mathbb{R}^3}{\text{minimize}} \quad & \frac{1}{2} \left\| H_k \tilde{v}_k^r - y_k \right\|_2^2 \\ \text{subject to} \quad & \tilde{v}_k^r \geq \left( \hat{a}_{f,k}^r - \gamma \right) \Delta t_r + \tilde{v}_{k-1}^r, \\ & \tilde{v}_k^r \leq \left( \hat{a}_{f,k}^r + \gamma \right) \Delta t_r + \tilde{v}_{k-1}^r. \end{aligned}$$

# Experiments (3/5) – Constraint

- Our idea visualization using real–world dataset, we used $\gamma = [5 \quad 5 \quad 4]^{\mathsf{T}}$.



# Methodology (4/5) – Bias Estimation

- The relationship between the estimated body velocity expressed in navigation frame $\hat{v}_b^n$ and the constrained ego-velocity $\tilde{v}_b^n$ can be established as[1]

$$\hat{v}_b^n = C_b^n C_r^b \tilde{v}^r - C_b^n \left[\bar{\omega}^b - \hat{b}_g\right]_\times p_r^b$$

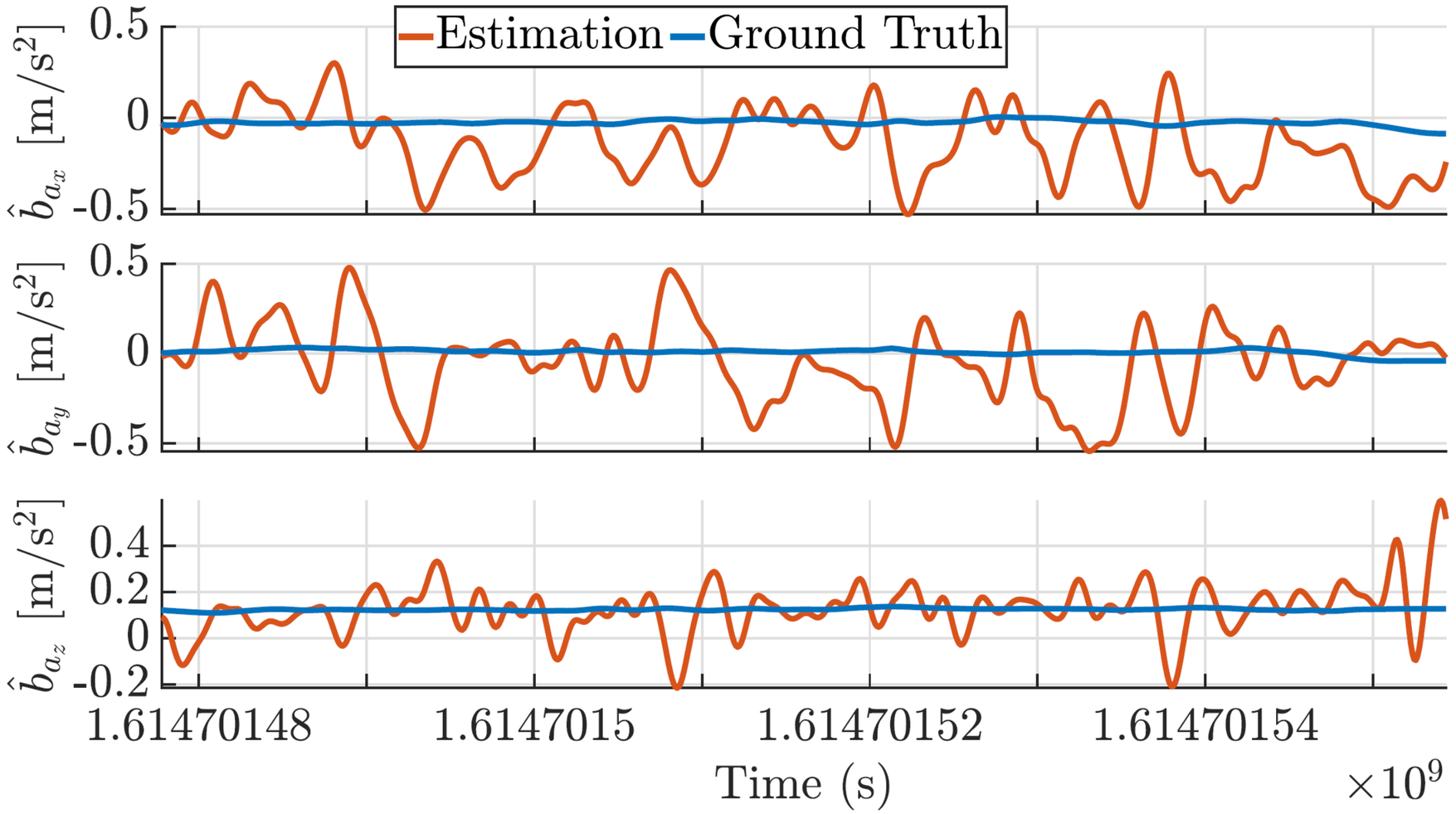
- We also can have

$$\hat{b}_a = \bar{f}^b + C_n^b \left( g^n - \frac{\hat{v}_{b,k}^n - \hat{v}_{b,k-1}^n}{\Delta t_r} \right)$$

[1]Do, Hoang Viet, et al. "DeRO: Dead Reckoning Based on Radar Odometry With Accelerometers Aided for Robot Localization." arXiv preprint arXiv:2403.05136 (2024).

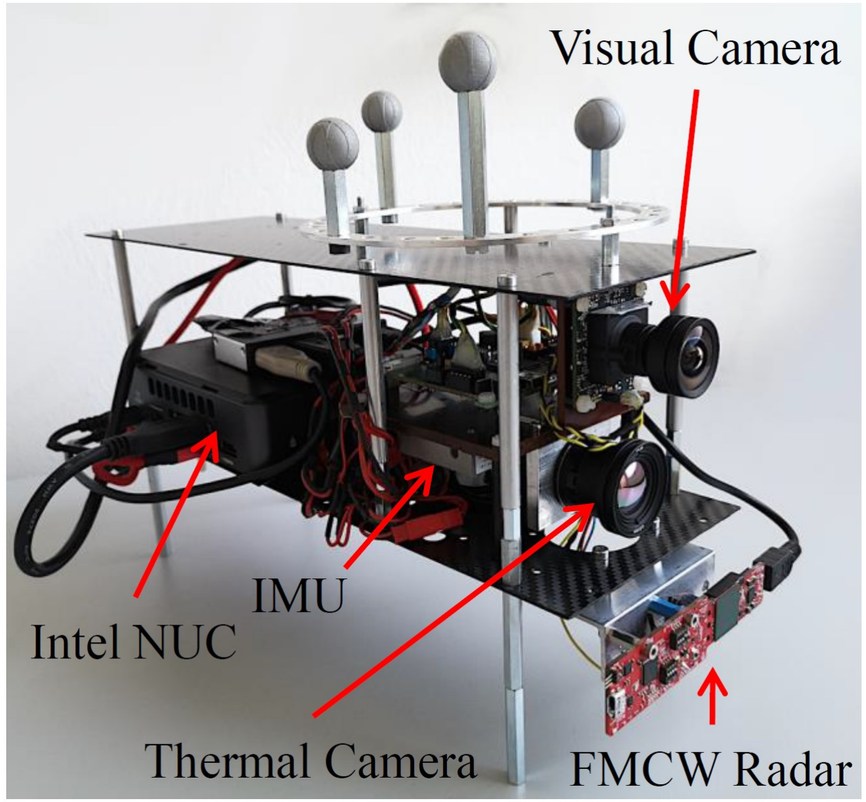
# Methodology (5/5) – Bias Estimation

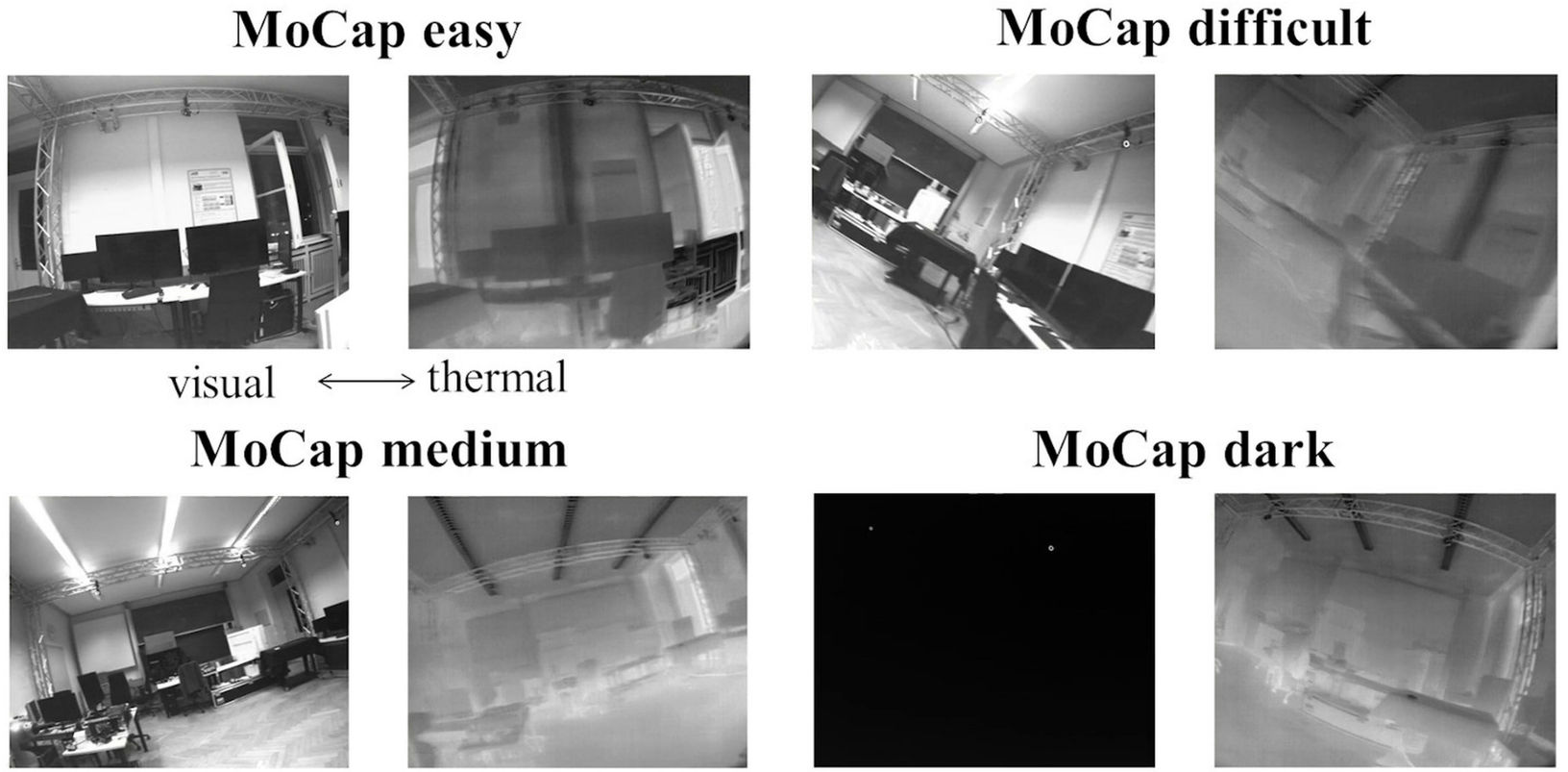
- Since accelerometer bias varies slowly over time, we intentionally apply a low-pass filter to smooth the outcomes (0.01 Hz).



# Experiments (1/5) – Sensor Platform

- To validate the proposed method, we used the renowned open-source IRS datasets[2] .
  - 4D FMCW radar TI IWR6843AOP (10Hz) and an IMU ADIS16448 (400Hz).

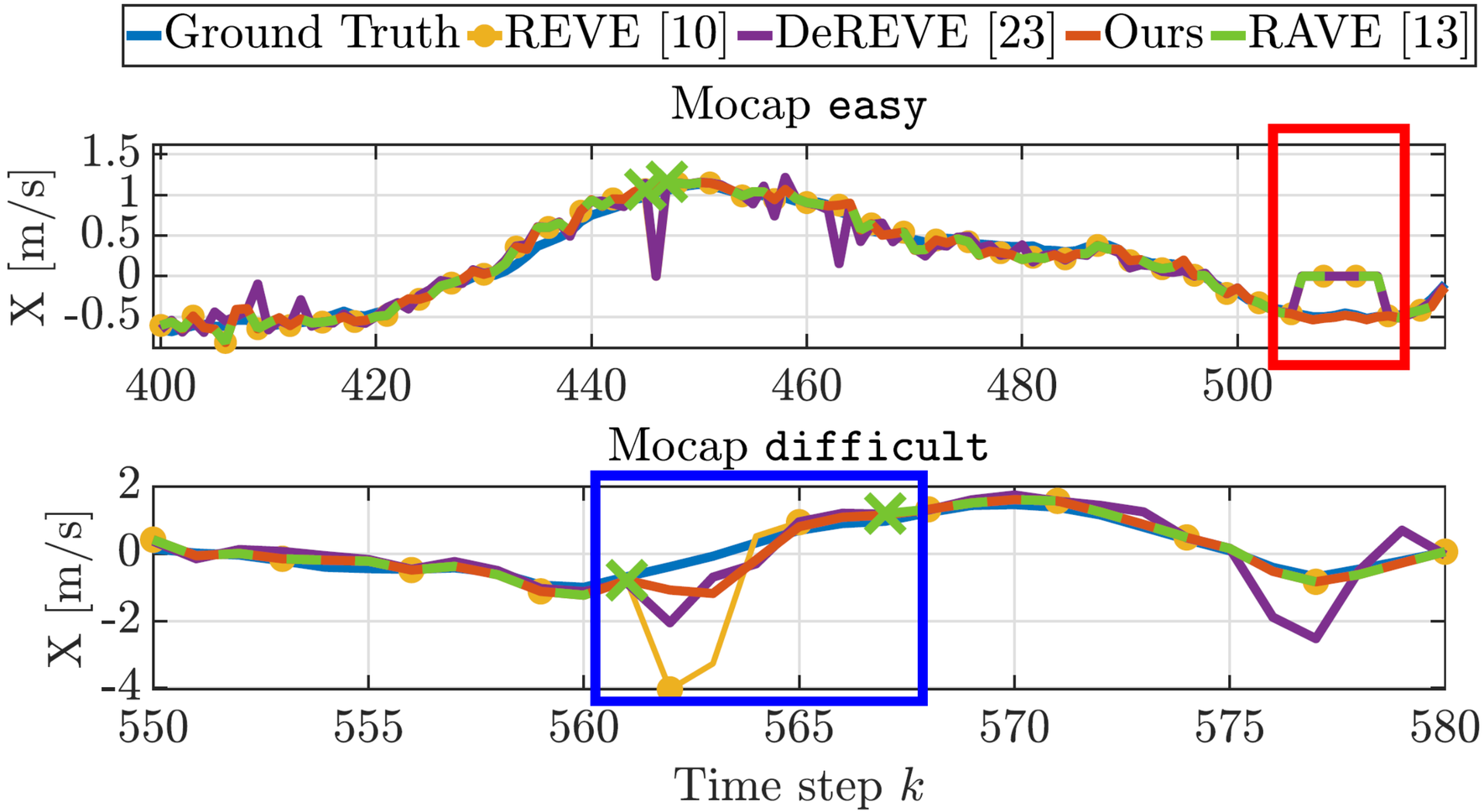



[2]C. Doer and G. F. Trommer, "Radar visual inertial odometry and radar thermal inertial odometry: Robust navigation even in challenging visual conditions," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 331–338.

# Experiments (2/5) – Estimation

- DeREVE displays some random peaks in its estimates.
- Conversely, RAVE consistently rejects anomalies, resulting in a discontinuous estimation.



# Experiments (3/5) – RMSE

- Compared to the conventional approach (REVE), the proposed method reduces the RMSE by approximately 36% in the x–axis, 51% in the y–axis, and 37% in the z–direction, on average.

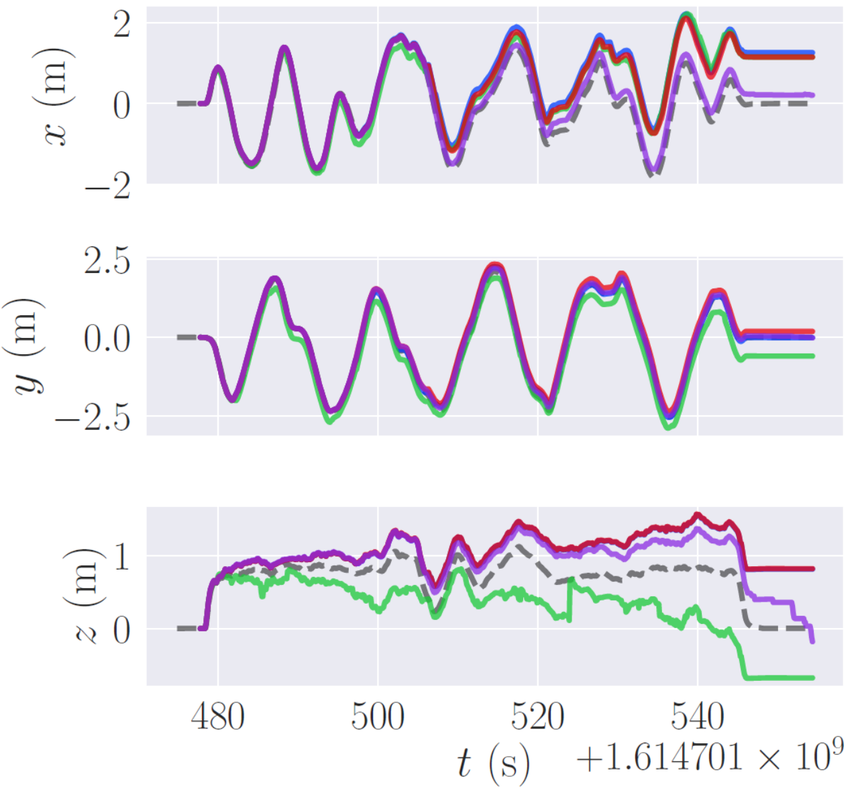
TABLE I: Experimental results for 3D ego-velocity estimation, presented as RMSE for each $x$-$y$-$z$ axis. The table also includes the average computation time across all datasets and the number of rejections by the RAVE method.

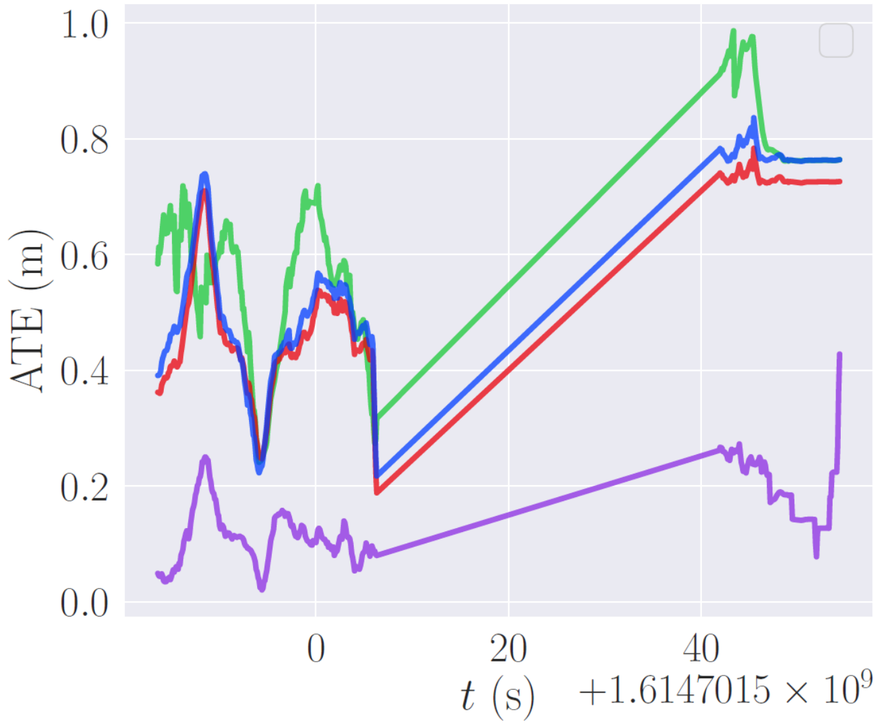
| Datasets | REVE [10] (**0.007** s/loop) | | | DeREVE [23] (0.013 s/loop) | | | RAVE [13] (**0.007** s/loop) | | | | Ours (0.008 s/loop) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $v_x$ | $v_y$ | $v_z$ | $v_x$ | $v_y$ | $v_z$ | $v_x$ | $v_y$ | $v_z$ | Reject | $v_x$ | $v_y$ | $v_z$ |
| Easy | 0.120 | 0.069 | **0.086** | 0.178 | 0.111 | 0.292 | 0.103 | 0.067 | **0.086** | 4/766 | **0.066** | **0.038** | 0.110 |
| Medium | 0.158 | 0.085 | 0.262 | 0.310 | 0.178 | 0.350 | 0.155 | 0.078 | 0.128 | 7/719 | **0.133** | **0.071** | **0.105** |
| Difficult | 0.307 | 0.269 | 0.319 | 0.652 | 0.490 | 0.840 | **0.151** | 0.101 | 0.178 | 30/721 | 0.157 | **0.090** | **0.171** |
| Dark | 0.153 | 0.080 | 0.132 | 0.389 | 0.177 | 0.449 | 0.138 | 0.079 | 0.122 | 8/1264 | **0.132** | **0.078** | **0.118** |
| Dark fast | 0.197 | 0.248 | 0.196 | 0.517 | 0.478 | 0.832 | 0.117 | **0.076** | 0.129 | 8/643 | **0.113** | 0.089 | **0.122** |
| Mean (m/s) | 0.187 | 0.150 | 0.199 | 0.409 | 0.287 | 0.553 | 0.133 | 0.080 | 0.129 | - | **0.120** | **0.073** | **0.125** |

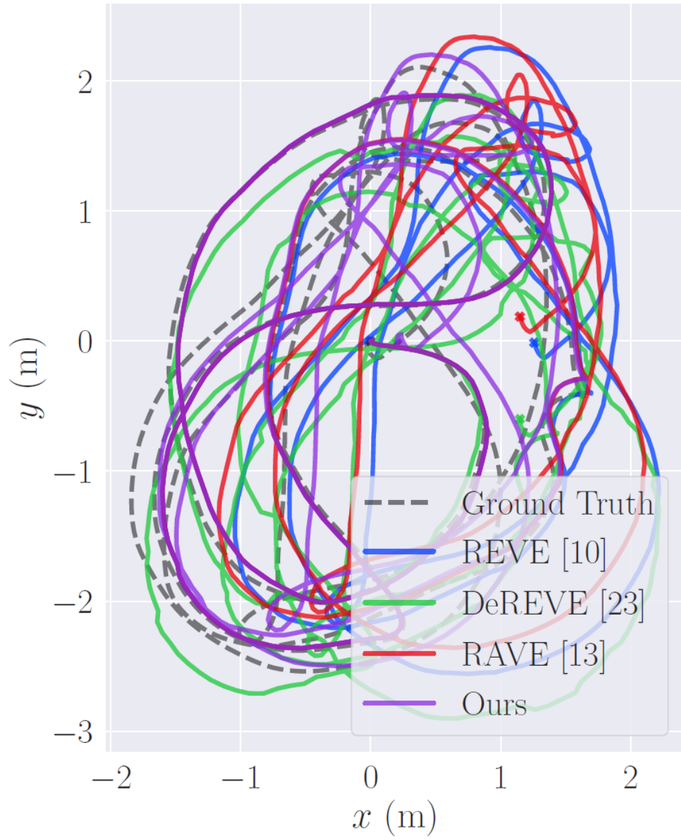
The bold numbers represent better results (the smaller number) and all values are rounded to three decimal digits.

# Experiments (4/5) – ATE

- Using evo toolboxes for ATE (Umeyama alignment) computation and analysis.




M. Grupp, “evo: Python package for the evaluation of odometry and slam.” *https://github.com/MichaelGrupp/evo*, 2017.

# Experiments (5/5) – ATE

- The proposed method reduces the ATE by approximately 53%, 84%, and 35% compared to REVE, DeREVE, and RAVE, respectively.

TABLE II: Absolute trajectory error calculated with *pos-yaw* alignment [32], presented in terms of translation errors across five datasets for all considered approaches.

| Datasets | REVE [10] | DeREVE [23] | RAVE [13] | Ours |
|---|---|---|---|---|
| Easy | 0.596 | 0.664 | 0.562 | **0.167** |
| Medium | 0.357 | 1.047 | 0.259 | **0.148** |
| Difficult | 0.646 | 2.726 | 0.483 | **0.281** |
| Dark | 0.339 | 0.959 | 0.289 | **0.271** |
| Dark Fast | 0.531 | 2.018 | **0.207** | 0.292 |
| Mean ATE (m) | 0.494 | 1.483 | 0.360 | **0.232** |

The bold numbers represent better results (the smaller number).
All values are rounded to three decimal digits.

[32] Z. Zhang and D. Scaramuzza, "A tutorial on quantitative trajectory evaluation for visual(-inertial) odometry," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

# Conclusion

- We propose
  - An acceleration-based **C**onstraint method for robust **R**adar **E**go-**V**elocity **E**stimation (CREVE). This framework functions as a submodule within a radar-inertial odometry system.
  - A practical bias estimation to enhance acceleration estimation from the accelerometer.
  - An adaptive parameter rule is designed to improve ego- velocity estimation accuracy.
- We conduct a comprehensive analysis against **three** existing state-of-the-art methods using the real-world dataset.

# Thank you!